

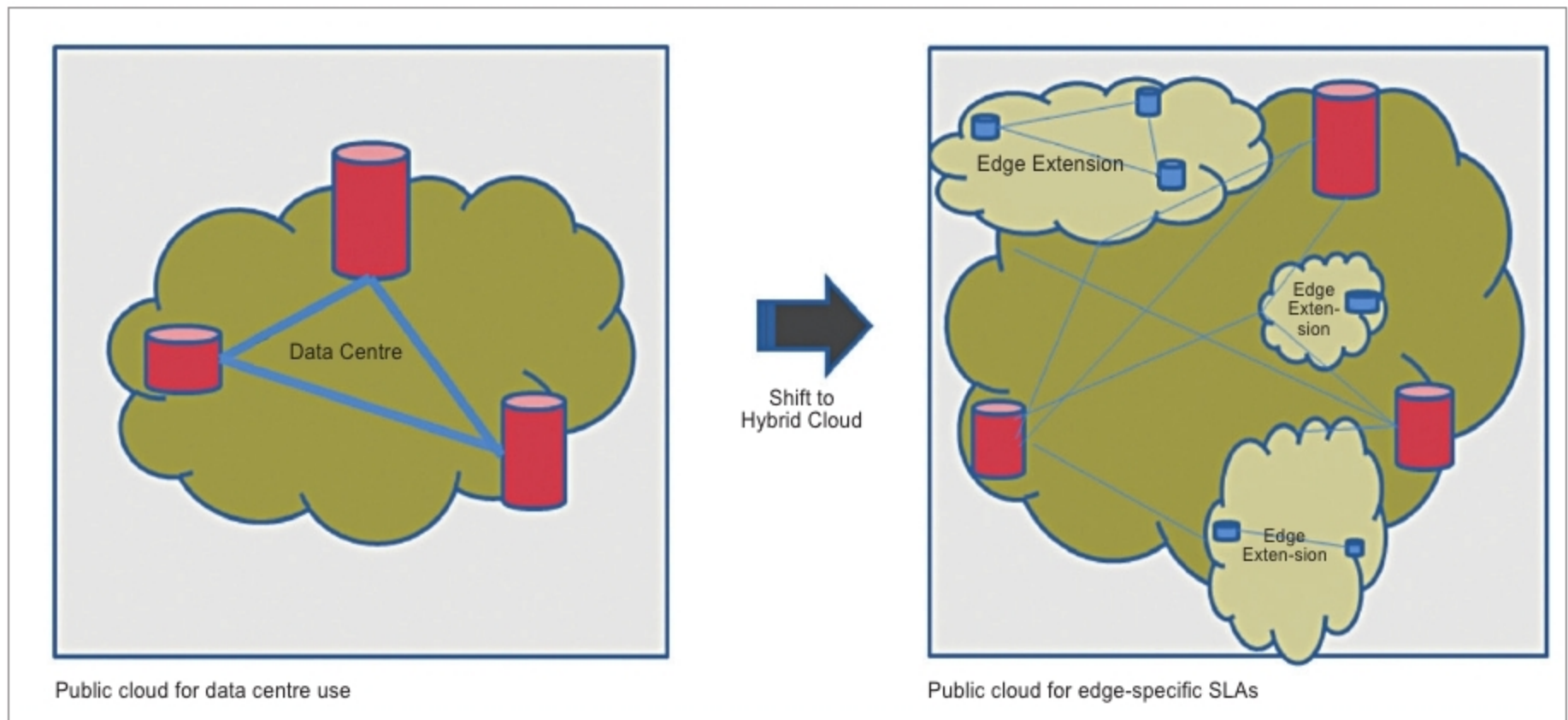


Figure 2: The shift to public cloud for edge-specific SLAs

Hybrid cloud brings in not only private cloud but public cloud as well. And even though edge is attributed to geographical location, it is well-understood that it is about a certain level of service-level agreements (SLAs) and key performance indicators (KPIs) required for specific application needs. Since the industry looks at hybrid cloud solutions, this approach feeds more fuel for edge.

Edge disrupts the cloud by not only providing a back office or central office but by also distributing the work at different levels, end-to-end, in terms of SLAs and KPIs.

In that sense, edge can be available at public cloud service providers. As for the many cloud service creations for small players of specific services on cloud, cloud service providers should offer the required SLAs needed for those specific services. This will help reduce the cost of infrastructure for certain services, which is marginal for such real-time requirements as telephony-based over-the-top (OTT) services. This can also help the small players to deploy their services without cost-intensive hardware and management.

This, in turn, puts pressure on cloud service providers to create their infrastructure in a way that addresses specific SLA demands, as these demands are generated by specific service providers

to perform as per the required KPIs. This could gain a fair extension in terms of cloud service providers' business portfolios, as shown in Figure 2.

This can also turn the business from virtualisation vendors or OEMs to cloud service providers, and re-define the term edge as cloud SLA or edge SLA instead of geographical earmarking.

What remains accessible from the geographical location point-of-view is fog. Fog is about the virtualisation of end system resources. These end systems are mutually connected through high bandwidth links and form a cluster. Fog allows an end system to share its scarce resources with others in a completely transparent way, and makes the entire cluster efficient for service demands.

Fog systems are end systems such as Wi-Fi access points or base transceiver systems in cellular networks. These are the first connection points for end users, and this is where they can provide real-time responses.

Fog systems can be better used for in-vehicle systems for autonomous or self-

driving applications. These can also be beneficial for such in-building solutions as shopping malls. Fog systems may also play an important role in 5G technology applications such as beam handovers.

As the networking business moves from OEMs to software vendors, many networking functions are now being developed on software and deployed on commodity servers. This has given a boost to virtual network functions (VNFs), which provide networking nodes functionalities such as routers and firewalls through software and can be run on commodity platforms.

Since cloud provides a data centre model for back office processing, these VNFs can also be deployed to the cloud. Also, the virtualisation technology helps manage cloud infrastructure in an optimum and efficient way, for the right use of computing resources as per the need and requirement of VNFs. It can also be configured on-the-go. This helps operators deploy their networking solutions in a flexible, scalable and cost-efficient way. **END** 🐧

By: Saurabh Verma

The author is founder-director and chief technology consultant, Fundarc Communication (xgnlab).

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